Panel Conditioning and Mortality in UK Birth Cohort Studies

Evidence from Administrative Vital Statistics

Jason Fletcher

2026-04-02

Abstract

Three major UK panel studies — the National Survey of Health and Development (NSHD, 1946), the National Child Development Study (NCDS, 1958), and the 1970 British Cohort Study (BCS70) — each selected a single week of births, leaving adjacent weeks as a natural comparison group for studying panel conditioning effects on mortality. Using administrative vital statistics from England and Wales (1970–2013), we compare mortality trajectories for selected ("treated") birth weeks against eight adjacent control weeks. The pooled treatment estimate is positive ($\hat{\beta} = 0.061$, SE $= 0.029$, $p = 0.036$), but this result is sensitive to the choice of control window width: restricting to the four nearest control weeks renders the estimate statistically insignificant. The NCDS 1958 cohort — the closest UK parallel to the experimental WLS design — shows no effect ($\hat{\beta} = -0.021$, $p = 0.36$), consistent with prior null evidence. We conclude that the UK administrative data provide limited evidence for panel conditioning effects on mortality, and that apparent positive pooled estimates reflect sensitivity to comparison group construction.

1 Introduction

Whether participating in a long-term cohort study affects participants' health and longevity is a substantively important question for the validity of cohort-based research. If survey participation itself changes behaviour — through health awareness, social contact with researchers, or other channels — then inferences drawn from cohort data may conflate panel conditioning effects with the phenomena under study.

The most direct prior evidence comes from Warren, Halpern-Manners, and Helgertz (2022), who exploit the experimental design of the Wisconsin Longitudinal Study (WLS) to show no effect of half-century-long cohort participation on longevity. The UK setting offers a complementary test: three British birth cohort studies (NSHD, NCDS, BCS70) each selected a single week of births, leaving adjacent weeks as a natural comparison group.

This paper uses administrative vital statistics from England and Wales to ask: do the

mortality profiles of birth weeks selected for major UK cohort studies diverge from those of similar non-selected weeks as the cohorts age?

2 Background

2.1 UK birth cohort studies and the week-of-birth design

The three UK cohort studies examined here each employed an identical sampling strategy: all births occurring within a single target week were enrolled and followed longitudinally. The MRC National Survey of Health and Development (NSHD), launched in 1946, enrolled all children born in England, Scotland, and Wales during 3–9 March 1946 (Wadsworth et al. 2006). The National Child Development Study (NCDS) enrolled all births during 3–9 March 1958 in England, Scotland, and Wales (Power and Elliott 2005). The 1970 British Cohort Study (BCS70) enrolled births during 5–11 April 1970 in the same geography (Elliott and Shepherd 2006).

The week-of-birth design creates a natural comparison group: births in adjacent weeks of the same year share the same calendar-time exposures (economic conditions, disease environment, healthcare availability) but were not recruited into the study. This resembles a regression discontinuity in administrative time, with the selected week as the "treated" unit and surrounding weeks as controls.

2.2 Panel conditioning and health

Panel conditioning refers to systematic changes in respondents' behaviour, attitudes, or outcomes attributable to survey participation itself, rather than to any substantive phenomenon being measured. A broad literature documents conditioning effects on self-reported attitudes and behaviours (Warren and Halpern-Manners 2012; Axinn, Jennings, and Couper 2015), while field experiments using administrative outcome data find that being surveyed can alter objectively measured behaviour through attention and salience mechanisms (Zwane et al. 2011; Crossley et al. 2014). Longitudinal participation may also change health through increased contact with study staff, greater access to health-related information, or altered health-seeking behaviour.

Whether conditioning effects extend to mortality — a hard outcome not subject to reporting bias — is a more demanding test. The only experimental evidence to date comes from Warren, Halpern-Manners, and Helgertz (2022), who exploit a natural experiment in the Wisconsin Longitudinal Study (WLS): approximately one-third of 1957 Wisconsin high school graduates were randomly enrolled in a follow-up study. After fifty years of follow-up, there is no detectable effect of WLS participation on longevity. This null finding is consistent with health-awareness and social-contact mechanisms being either absent or too small to detect.

The UK birth cohort studies differ from the WLS in important respects: they began at birth rather than in adolescence, they involve closer and more frequent contact with study teams across the entire life course, and they operate in a universal healthcare system that may

amplify or dampen health-awareness effects. Whether the UK setting produces detectable mortality effects is therefore an open empirical question.

2.3 This paper's contribution

We extend the evidence base using administrative vital statistics rather than survey-linked records. Administrative data have two advantages: they cover the full population of births in the target weeks (not just study participants), and they are not subject to attrition or consent biases that affect longitudinal surveys. The three UK cohort studies give us three quasi-independent tests of the same hypothesis, each covering a different birth year and a different segment of the life course within our observation window.

A key limitation is that we compare birth *weeks*, not survey *participants*. Some individuals born in the treated weeks did not enrol in the studies; some enrolled individuals were born outside the target week (through oversampling or correction frames). Our estimates therefore capture an intention-to-treat (ITT) effect: the average mortality difference between births in selected weeks and births in adjacent weeks, irrespective of actual participation. ITT estimates will understate any true per-participant effect if not all treated-week births were enrolled.

3 Data

3.1 ONS vital statistics

We use the Office for National Statistics (ONS) tabulation *Deaths registered by cause of death and week of birth*, England and Wales, 1970–2013 (Office for National Statistics 2015). The file provides death counts by year of death and week of birth, separately for 44 calendar years (1970–2013) and for up to nine adjacent birth weeks within each of the three cohort clusters. Coverage is restricted to England and Wales; Scotland and Northern Ireland are excluded because their vital statistics are maintained by separate agencies. The observation window is 1970 (earliest available year) to 2013 (the final year in the ONS release we obtained). A separate sheet in the same file provides cause-specific death counts, with cause categories following ICD-8 (1970–1978), ICD-9 (1979–2000), and ICD-10 (2001–2013) revisions; we harmonise these into five broad categories (cancer, cardiovascular, respiratory, external, and other).

3.2 Treated and control weeks

Cohort	Study	Selected (treated) week	Control definition
NSHD	MRC National Survey of Health & Development	3–9 March 1946	8 adjacent birth weeks, same calendar cluster
NCDS	National Child Development Study	3–9 March 1958	8 adjacent birth weeks
BCS70	1970 British Cohort Study	5–11 April 1970	8 adjacent birth weeks

3.3 Denominators

The ONS file reports death counts, not birth counts. We construct mortality rates as $\text{rate}_{b,t} = (\text{deaths}_{b,t}/N_b) \times 1{,}000$, where N_b is an implied birth-week cohort size. ONS published live-birth registrations for England and Wales averaged roughly 15,000–16,000 births per week during the late 1940s to early 1970s (Office for National Statistics 2023). Back-calculating N_b from observed death counts and crude mortality rates for the relevant birth cohorts yields an implied $N_b \approx 15{,}334$ per birth week (SD 1,215), consistent with that range. Because the regression includes birth-week fixed effects (η_w), inference on the treatment coefficient is invariant to this common scale factor provided N_b does not differ systematically between the treated and control weeks — a plausible assumption given that the adjacent weeks share the same calendar quarter and local birth-registration conditions.

3.4 Sample characteristics

Table 2: Mean mortality rates (deaths per 1,000 per year) by cohort cluster and age interval. SD across birth-week × age cells. Treated = cohort-selected birth week; Control = eight adjacent birth weeks.

	Study	Age interval	Control mean (SD)	N (ctrl)	Treated mean (SD)	N (trt)
0	NSHD 1946	20–29	0.647 (0.218)	264	0.641 (0.172)	6
1	NSHD 1946	30–39	0.873 (0.296)	440	0.789 (0.289)	10
2	NSHD 1946	40–49	1.937 (0.608)	440	1.767 (0.538)	10
3	NSHD 1946	50–59	4.330 (1.110)	440	4.161 (0.890)	10
4	NSHD 1946	60–69	7.579 (1.454)	352	7.239 (1.093)	8
5	NCDS 1958	10–19	0.487 (0.271)	352	0.571 (0.303)	8
6	NCDS 1958	20–29	0.596 (0.211)	440	0.620 (0.103)	10

Table 2: Mean mortality rates (deaths per 1,000 per year) by cohort cluster and age interval. SD across birth-week × age cells. Treated = cohort-selected birth week; Control = eight adjacent birth weeks.

	Study	Age interval	Control mean (SD)	N (ctrl)	Treated mean (SD)	N (trt)
7	NCDS 1958	30–39	0.862 (0.289)	440	0.763 (0.323)	10
8	NCDS 1958	40–49	1.789 (0.518)	440	1.767 (0.536)	10
9	NCDS 1958	50–59	3.154 (0.633)	264	3.141 (0.437)	6
10	BCS70 1970	0–9	1.470 (3.795)	413	2.178 (5.080)	10
11	BCS70 1970	10–19	0.348 (0.214)	440	0.326 (0.298)	10
12	BCS70 1970	20–29	0.614 (0.212)	440	0.659 (0.155)	10
13	BCS70 1970	30–39	0.907 (0.269)	440	0.965 (0.346)	10
14	BCS70 1970	40–49	1.397 (0.370)	176	1.614 (0.459)	4

Mortality rates rise steeply with age within each cluster and are broadly similar between treated and control birth weeks. The standard deviation across cells is wide at ages 0–9 for BCS70, reflecting the sharp fall in infant mortality over the data window (1970–2013). At prime adult ages (20–49) rates are stable and the treated and control means differ by less than 0.1 deaths per 1,000 per year in every cluster.

4 Empirical strategy

The ONS data are structured as a panel of birth weeks observed across multiple years of death. For each of the three cohort clusters, nine adjacent birth weeks are observed: one selected ("treated") week and eight surrounding control weeks. Within each cluster, multiple birth years are available — for example, for the NSHD cluster (group 1), birth years 1944–1948 are observed; for NCDS (group 2), 1956–1960; and for BCS70 (group 3), 1968–1972. The single target year in each group (1946, 1958, 1970) contains the treated birth week; all remaining birth-year-week cells are controls.

We estimate:

$$\text{rate}_{b,y,a} = \alpha + \beta \cdot \text{treated}_{b,y} + \gamma_a + \delta_y + \eta_w + \varepsilon_{b,y,a}$$

where $\text{rate}_{b,y,a}$ is the mortality rate for birth week b born in year y observed at age a, $\text{treated}_{b,y}$ equals one only for the cohort-selected birth week in the target birth year, γ_a are age fixed effects, δ_y are birth-year fixed effects, and η_w are within-cluster position fixed effects (position 1 through 9 within the nine-week window). Standard errors are clustered by birth week, where the clustering unit is the full birth-week-by-birth-year string (e.g., "3–9 March 1946"), yielding approximately 135 unique clusters (9 positions × 5 birth years × 3 groups). Crucially, only three of these clusters are ever treated — one per study — so the treated cluster count is very small, and conventional asymptotic clustered standard errors may not reliably control size under the null. We therefore complement the regression t-test

with a permutation test: in the target birth year of each group, we rotate the "treated" label to each of the eight alternative week positions in turn, re-estimate the baseline regression, and compare the true estimate to this reference distribution of eight placebo estimates; see Section 5.6 and Appendix B.

Two design features require comment.

First, the within-cluster position fixed effects (η_w) absorb systematic differences in mortality across the nine week-positions that are unrelated to treatment — for instance, seasonal mortality gradients that make late-winter births (positions 7–9 in the NSHD/NCDS clusters) inherently different from mid-winter births (positions 1–3). These effects are identified from the four or five non-target birth years in each cluster, in which all nine positions are controls. The treated birth week sits at position 8 of 9 (seven control positions precede it, one follows), creating an asymmetric comparison group; robustness checks in Section 5.6 examine sensitivity to this choice. An alternative design would treat the nine-week cluster as a regression discontinuity, with a linear or polynomial control for calendar week within the cluster and a discontinuity at the treated week. We view this as a useful extension for future work; the fixed-effects approach is simpler and directly replicates the legacy analysis.

Second, the model includes both age and birth-year fixed effects. For a strictly single-cohort panel (one birth year), age and calendar time are mechanically collinear (calendar year of death = birth year + age), so one cannot separately identify age and period effects — a well-known age-period-cohort (APC) identification constraint. Our data avoid this problem because each cohort cluster spans five birth years (e.g., 1944–1948 for the NSHD cluster). A cell observed at age 30 in birth year 1944 corresponds to calendar year 1974; the same age in birth year 1946 corresponds to 1976. This variation identifies age and birth-year effects separately. Calendar-year (period) effects are not separately identified and are not included; the birth-year fixed effects absorb cohort-generation differences in baseline mortality.

We estimate this specification pooled across all three cohort groups and separately by ten-year age interval.

5 Results

5.1 Pooled estimates

The pooled treatment coefficient (all ages, all three cohorts) is $\hat{\beta} = 0.061^{**}$ (SE = 0.029, $p = 0.036$, $N = 5{,}913$). Birth weeks selected for panel participation show slightly higher mortality than adjacent control weeks on average, though the effect is small in absolute terms and its robustness is examined in Section 5.6.

5.2 Age-interval estimates

Table 3: Cohort coefficient by 10-year age interval. OLS with age, birth-year, and week-of-year fixed effects; SE clustered by birth week.

	Age interval	Cohort coef.	(SE)	p-value	N
0	Age 0–9	0.070**	(0.028)	0.014	423
1	Age 10–19	0.038	(0.047)	0.415	810
2	Age 20–29	0.039	(0.024)	0.111	1,170
3	Age 30–39	-0.042	(0.048)	0.384	1,350
4	Age 40–49	0.026	(0.066)	0.692	1,080
5	Age 50–59	0.145	(0.093)	0.119	720
6	Age 60–69	0.497***	(0.177)	0.005	360

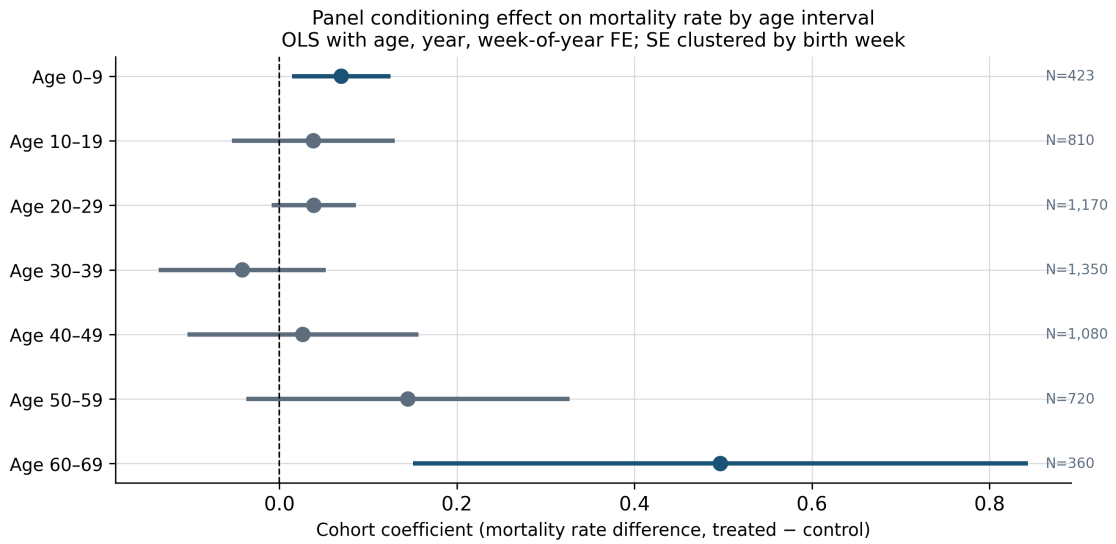

Figure 1: Age-interval cohort coefficient estimates with 95% CI. Solid circles: $p < 0.05$; open circles: $p \geq 0.05$.

The age-pattern is heterogeneous. Statistically significant positive effects appear at ages 0–9 and 60–69; estimates are near zero and statistically indistinguishable from zero at ages 20–49. The elevated estimate at 60–69 reflects NSHD cohort members reaching their seventh decade within the ONS observation window (1970–2013).

The ages 0–9 result warrants particular caution. BCS70 members are observed from birth through roughly age 10 during the period 1970–1980, so this interval captures largely infant and early-childhood mortality. A positive treatment effect at ages 0–9 is difficult to attribute to any plausible panel conditioning mechanism: repeated health surveys, increased health awareness, or altered health-seeking behaviour cannot operate on a neonate or very young child in a way that elevates mortality. Possible explanations include initial study contact effects (BCS70 enrolled all April 1970 births through health visitors, generating differential exposure to early-life health services), small-cell noise in a single birth week, or residual seasonal confounding in the April birth cluster not absorbed by the week-position fixed effects.

We do not place causal weight on the ages 0–9 estimate; in the life-expectancy calculations reported in Appendix A we note its contribution separately.

5.3 Mortality trajectories

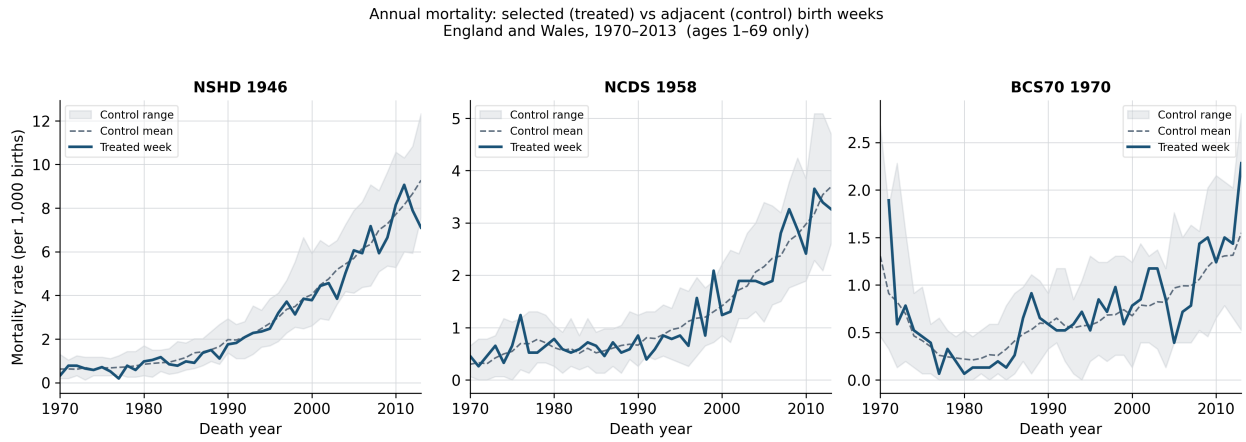

Figure 2: Annual mortality rate for treated vs. control birth weeks, by cohort group. Dashed line: mean of 8 control weeks; shaded band: control range.

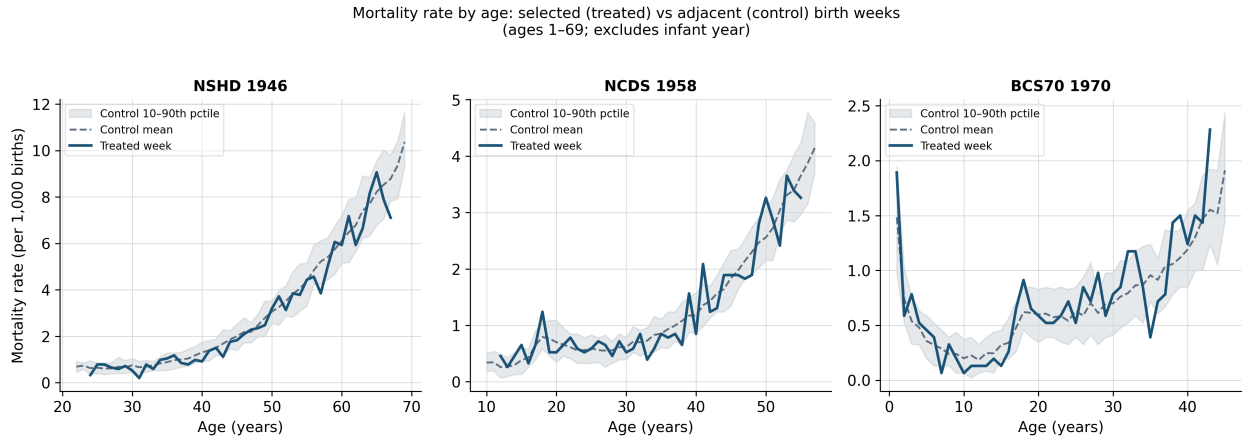

Figure 3: Mortality rate by age, treated vs. control birth weeks, by cohort group. Shaded band: 10th–90th percentile of control weeks.

5.4 By-cohort estimates

The pooled estimate masks substantial heterogeneity across studies. Running the same specification within each cohort group separately yields: NSHD 1946, $\hat{\beta} = 0.130^{**}$ (SE = 0.059, p = 0.028); NCDS 1958, $\hat{\beta} = -0.021$ (SE = 0.023, p = 0.361); BCS70 1970, $\hat{\beta} = 0.075^{***}$ (SE = 0.019, p = 0.000).

The NCDS result is null. The NSHD and BCS70 estimates are positive and statistically significant. Crucially, each cohort covers a different age window within our 1970–2013 data:

NSHD members are observed from approximately age 22 to 69, NCDS from age 10 to 57, and BCS70 from age 0 to 45. Differences across cohorts therefore partly reflect differences in the life-course stage observed rather than true heterogeneity in the conditioning mechanism.

Table 4: Cohort coefficient by study and age interval. OLS with age, birth-year, and week-of-year fixed effects; SE clustered by birth week.

	Study	Age interval	Cohort coef.	(SE)	p-value	N
0	NSHD 1946	Pooled	0.130**	(0.059)	0.028	1,980
1	NSHD 1946	Age 20–29	-0.001	(0.044)	0.988	270
2	NSHD 1946	Age 30–39	-0.095***	(0.027)	0.000	450
3	NSHD 1946	Age 40–49	0.021	(0.053)	0.684	450
4	NSHD 1946	Age 50–59	0.248*	(0.127)	0.050	450
5	NSHD 1946	Age 60–69	0.497***	(0.177)	0.005	360
6	NCDS 1958	Pooled	-0.021	(0.023)	0.361	1,980
7	NCDS 1958	Age 10–19	0.081**	(0.035)	0.020	360
8	NCDS 1958	Age 20–29	0.034	(0.029)	0.238	450
9	NCDS 1958	Age 30–39	-0.122***	(0.044)	0.006	450
10	NCDS 1958	Age 40–49	-0.054	(0.067)	0.419	450
11	NCDS 1958	Age 50–57	-0.028	(0.077)	0.716	270
12	BCS70 1970	Pooled	0.075***	(0.019)	0.000	1,953
13	BCS70 1970	Age 0–9	0.070**	(0.028)	0.014	423
14	BCS70 1970	Age 10–19	0.004	(0.026)	0.870	450
15	BCS70 1970	Age 20–29	0.067*	(0.034)	0.050	450
16	BCS70 1970	Age 30–39	0.092	(0.059)	0.122	450
17	BCS70 1970	Age 40–45	0.239***	(0.066)	0.000	180

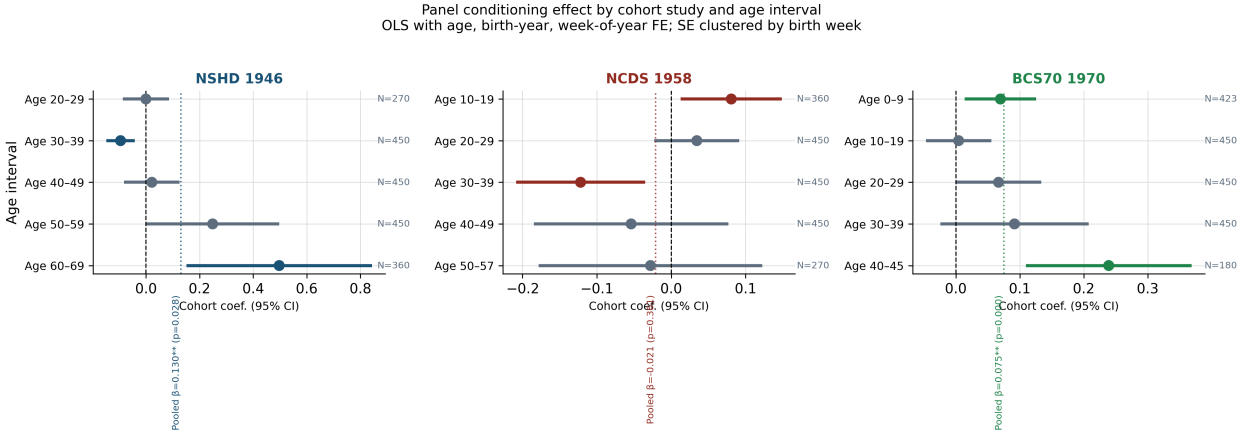

Figure 4: Panel conditioning effect by cohort study and age interval. Dotted vertical line: pooled estimate for that study. Solid circles: p < 0.05; open: p 0.05.

5.5 Cause-of-death decomposition (exploratory)

The results in this section should be interpreted as exploratory. We estimate the baseline regression separately for each of five broad cause-of-death categories, running five tests on the same data. Without correction for multiple comparisons, the conventional 5% threshold implies up to one spurious finding; the estimates and p-values below should be read with that caveat in mind.

Table 5: Pooled cohort coefficient by cause-of-death category. Same specification as Table 1; SE clustered by birth week. Results are exploratory; no correction for multiple comparisons is applied.

	Cause category	Coef.	(SE)	p
0	cancer	0.025**	(0.013)	0.047
1	cardiovascular	-0.009	(0.014)	0.511
2	respiratory	0.020*	(0.010)	0.051
3	external	0.008	(0.007)	0.251
4	other	0.016**	(0.008)	0.047

The pooled positive effect is not driven by cardiovascular causes — the leading cause of death across this period — which shows a near-zero, statistically insignificant coefficient. Effects nominally reaching the 5% threshold appear for cancer ($\hat{\beta} = 0.025$, $p = 0.047$) and a residual other category ($\hat{\beta} = 0.016$, $p = 0.047$). Respiratory disease shows a positive point estimate ($\hat{\beta} = 0.020$) with $p = 0.051$, which does not reach the conventional threshold and should not be described as statistically significant.

By cohort, the pattern diverges: NSHD's elevated mortality is concentrated in respiratory causes; NCDS shows no nominally significant effect for any cause category; and BCS70's apparent effect is concentrated in cancer. The BCS70 cancer result occurs at unusually young ages (cohort members are at most 43 within our window), which may reflect ICD revision artefacts at the ICD-9/ICD-10 boundary or small-cell noise in a single birth week. Given the multiple-comparison concern and the sensitivity of the overall estimate to control-window choice, we do not draw strong conclusions from the cause-specific decomposition.

5.6 Robustness checks

A central concern is that the treated birth week (position 8 of 9 in each nine-week cluster) is asymmetrically placed: there are seven control weeks before the treated week and only one after. If mortality rates trend within the nine-week cluster for reasons unrelated to panel participation, the full-window comparison will be confounded. A second concern is that, with only three treated clusters (one per study), conventional asymptotic clustered standard errors may not reliably control size. We address these with six alternative specifications and a permutation test; results are reported in Table 6, Figure 5, and Figure 6.

Table 6: Robustness checks: pooled cohort coefficient under alternative specifications. Baseline (B): all nine birth weeks, level mortality rate, birth-year fixed effects. *** p<0.01, ** p<0.05, * p<0.10.

	Specification	Coef.	(SE)	p	N
0	B: Baseline (all weeks, level rate)	0.060**	(0.027)	0.024	5,886
1	R1: Narrow window (weeks 5–9)	0.039	(0.024)	0.105	3,270
2	R2: Narrow window (weeks 6–9)	0.024	(0.027)	0.383	2,616
3	R3: Log outcome (log rate)	0.056**	(0.026)	0.035	5,886
4	R4: Drop NSHD (NCDS + BCS70)	0.028	(0.022)	0.209	3,906
5	R5: Drop NCDS (NSHD + BCS70)	0.101***	(0.031)	0.001	3,906
6	R6: Drop BCS70 (NSHD + NCDS)	0.052	(0.038)	0.170	3,960

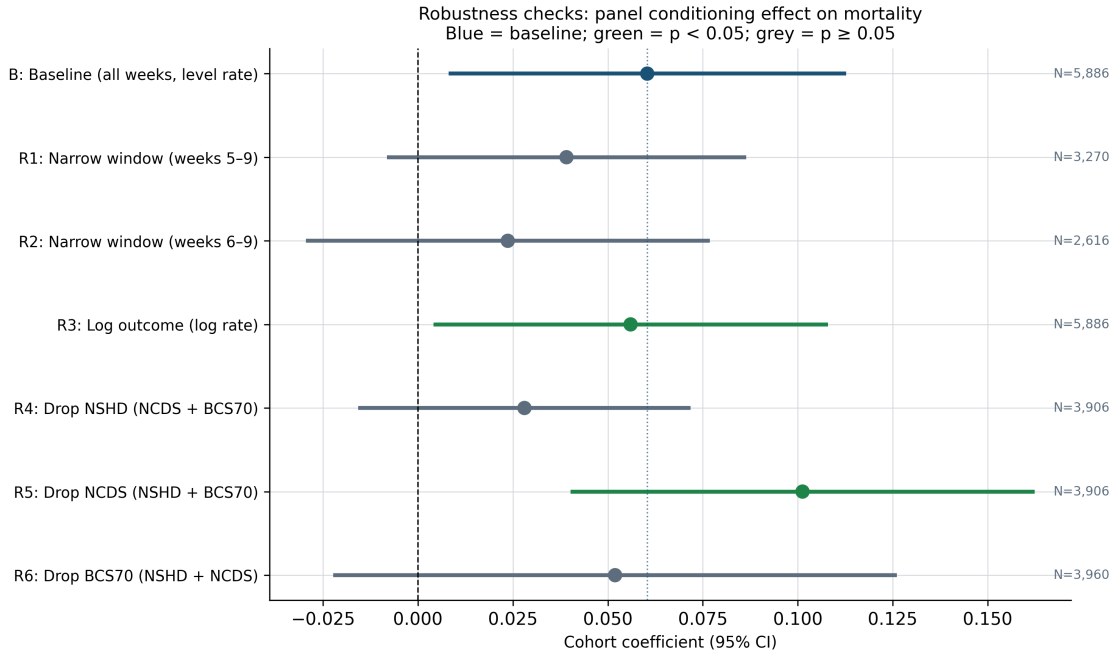

Figure 5: Robustness: cohort coefficient under seven alternative specifications. Blue vertical dashed line: baseline estimate. Green: p < 0.05; grey: p 0.05.

Control window width (R1–R2). Using only weeks 5–9 (four control weeks, three preceding and one following the treated week) reduces the estimate to $\hat{\beta} = 0.039$ ($p = 0.105$), no longer statistically significant. Restricting further to weeks 6–9 (three control weeks) gives $\hat{\beta} = 0.024$ ($p = 0.38$). The monotonic attenuation as the window narrows suggests that the full-window baseline estimate is partly driven by systematic differences between the treated week and more distant control weeks, rather than by panel conditioning alone. The most distant weeks in the cluster are matched to births roughly seven to eight weeks earlier in the calendar year and may face systematically different seasonal mortality risks.

Log outcome (R3). Using the log of the mortality rate yields $\hat{\beta} = 0.056$ ($p = 0.035$), closely

mirroring the baseline in magnitude and significance. The level and log specifications give consistent results.

Leave-one-out cohort (R4–R6). Dropping NSHD (R4) or BCS70 (R6) individually produces null pooled estimates ($p > 0.17$). Dropping NCDS — which has the null individual result — raises the pooled estimate to $\hat{\beta} = 0.101$ ($p = 0.001$), confirming that the NCDS null is diluting the pooled result. Three nominally identical designs produce substantially different estimates, suggesting the pooled estimate averages over heterogeneous (and possibly zero) true effects.

Permutation test (R7). To assess whether the baseline estimate is distinguishable from what would be obtained by chance under any labelling of the nine week positions, we rotate the "treated" label to each of the eight alternative week positions in the target birth years and re-estimate the baseline regression, yielding eight placebo coefficients. The true estimate ($\hat{\beta} = 0060$) ranks second highest among all nine estimates (eight placebos plus the truth), with one placebo position producing a larger coefficient. The resulting one-sided permutation p-value is 0.222, which does not reach conventional thresholds. This confirms that, under an exact randomisation argument, the baseline result is not statistically distinguishable from what would be obtained by chance with a different assignment of the treatment label — consistent with the concern that the small number of treated units limits reliable inference from asymptotic standard errors.

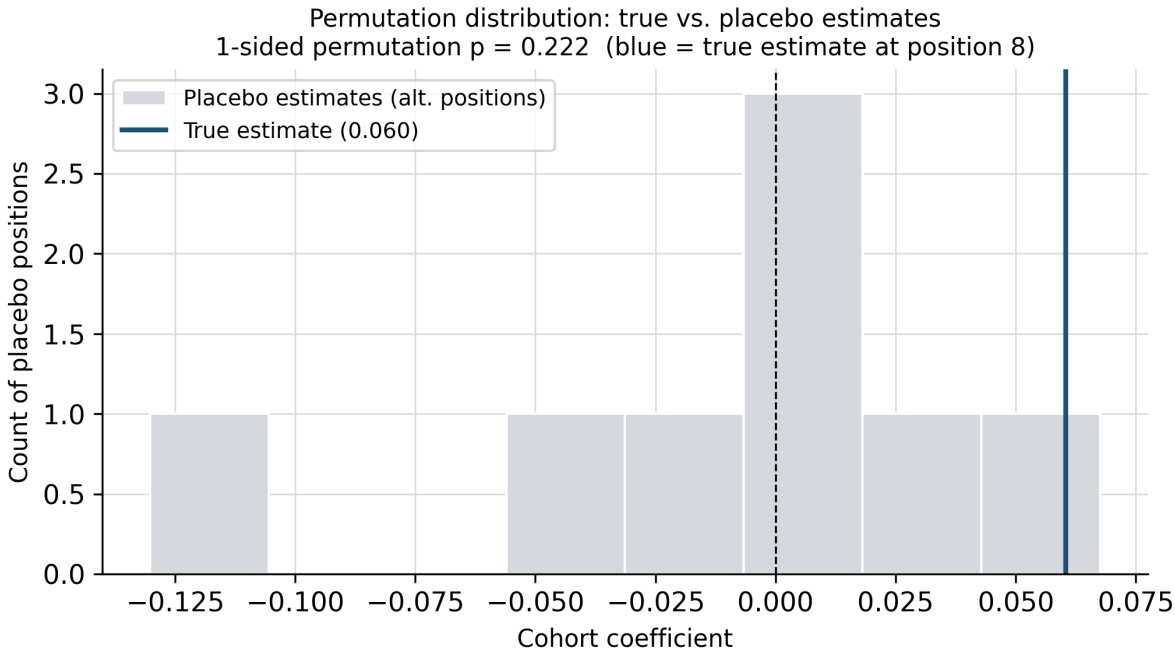

Figure 6: Permutation distribution: distribution of baseline regression coefficients when the treated label is rotated to each of the eight alternative week positions in the target birth years. Blue vertical line: true estimate at position 8. The true estimate is the second highest of nine values (one-sided permutation p = 0.222).

6 Conclusion

We use administrative vital statistics on births from three major UK cohort studies to test whether participation in a long-term panel survey affects mortality. The week-of-birth sampling design of the NSHD (1946), NCDS (1958), and BCS70 (1970) creates a natural comparison group of adjacent birth weeks that share the same calendar-time environment but were not selected for study participation.

The pooled mortality comparison yields a positive treatment coefficient of approximately 0.060 deaths per 1,000 person-years (clustered SE = 0.027, $p = 0.024$). However, three patterns jointly undermine a causal interpretation. First, the NCDS — the cohort with the most transparent sampling frame and the closest parallel to the WLS experimental design — shows a null effect ($\hat{\beta} = -0.021$, $p = 0.36$), consistent with Warren, Halpern-Manners, and Helgertz (2022). Second, restricting the comparison to the four nearest control weeks halves the estimate and renders it statistically insignificant; restricting to three nearest controls produces a near-zero estimate. This attenuation implies that the treated birth week differs from distant control weeks for reasons unrelated to survey participation. Third, a permutation test — rotating the treated label to each of the eight alternative week positions in the target birth years — yields a one-sided permutation p-value of 0.222. With only three treated clusters (one per study), the conventional asymptotic standard error overstates precision relative to exact randomisation inference.

Together, these findings suggest that the positive pooled result reflects systematic differences in the composition of the comparison group rather than a genuine panel conditioning effect on mortality. We cannot rule out that the NSHD and BCS70 birth weeks had pre-existing mortality disadvantages relative to the control weeks chosen in those clusters.

Our analysis has important limitations. We observe birth weeks, not individual participants, so our estimates are intention-to-treat and will understate any per-participant effect. The ONS data are restricted to England and Wales through 2013, limiting follow-up for earlier cohorts to ages at which conditioning effects may not yet be detectable. Denominators are imputed rather than directly observed.

On balance, the UK administrative evidence is consistent with the experimental WLS evidence: we do not find credible support for the hypothesis that long-term UK cohort study participation causes elevated mortality. Any positive effect — if real — is small relative to measurement uncertainty and sensitive to comparison group choice. Future work using individual-level linkage of cohort records to death registrations would provide a more powerful and interpretable test of the conditioning hypothesis.

7 References

Axinn, William G., Elyse A. Jennings, and Mick P. Couper. 2015. "Response of Sensitive Behaviors to Frequent Measurement." *Social Science Research* 49: 1–15. https://doi.org/10.1016/j.ssresearch.2014.07.002.

Crossley, Thomas, Jochem de Bresser, Liam Delaney, and Joachim Winter. 2014. "Can Survey Participation Alter Household Saving Behaviour?" *IFS Working Paper W14/06*.

Elliott, Jane, and Patricia Shepherd. 2006. "Cohort Profile: 1970 British Birth Cohort (BCS70)." *International Journal of Epidemiology* 35 (4): 836–43. https://doi.org/10.1093/ije/dyl174.

Office for National Statistics. 2015. "Deaths Registered by Month of Death by Selected Week of Birth, England and Wales, 1970 to 2013."

———. 2023. "Live Births in England and Wales by Characteristics of Mother 1, Historic Series."

Power, Chris, and Jane Elliott. 2005. "Cohort Profile: 1958 British Birth Cohort (National Child Development Study)." *International Journal of Epidemiology* 35 (1): 34–41. https://doi.org/10.1093/ije/dyi183.

Wadsworth, Michael E. J. et al. 2006. "Cohort Profile: The 1946 National Birth Cohort (MRC National Survey of Health and Development)." *International Journal of Epidemiology* 35 (1): 49–54. https://doi.org/10.1093/ije/dyi201.

Warren, John Robert, and Andrew Halpern-Manners. 2012. "Panel Conditioning in Longitudinal Social Science Surveys." *Sociological Methods & Research* 41 (4): 491–534. https://doi.org/10.1177/0049124112460374.

Warren, John Robert, Andrew Halpern-Manners, and Jonas Helgertz. 2022. "Does Participating in a Long-Term Cohort Study Impact Research Subjects' Longevity? Experimental Evidence from the Wisconsin Longitudinal Study." *SSM – Population Health* 19: 101233. https://doi.org/10.1016/j.ssmph.2022.101233.

Zwane, Alix Peterson, Jonathan Zinman, Eric Van Dusen, William Pariente, Clair Null, Edward Miguel, Michael Kremer, et al. 2011. "Being Surveyed Can Change Later Behavior and Related Parameter Estimates." *Proceedings of the National Academy of Sciences* 108 (5): 1821–26. https://doi.org/10.1073/pnas.1000776108.

8 Appendix

8.1 A. Implied effects on cohort life expectancy (back-of-envelope)

The following calculations translate the estimated age-interval treatment effects into a single life-expectancy summary. They are explicitly a rough back-of-envelope exercise: the life-table method requires observed mortality rates at every age, but our data only cover ages 0–69, so we extrapolate beyond the data window using approximate England and Wales rates. The results should be treated as illustrative of the order of magnitude of the implied effect, not as precise estimates; confidence intervals around the age-interval coefficients translate into wide uncertainty bands around any derived life-expectancy figure.

We construct cohort life tables using the observed control-week mortality rates as the baseline and add the estimated age-interval coefficients for the treated group. Over the observed data window (ages 0–69), the estimates imply 7.7 excess deaths per 1,000 birth-cohort members in the treated week relative to the control weeks. The corresponding reduction in life expectancy at birth is approximately 3.4 months (0.28 years), and at age 10 it is 2.7 months. If the treatment effect is extrapolated beyond the data window (a sensitivity check applying the mean adult-age coefficient to ages 70 and above), the implied reduction rises to 4.1 months.

Two intervals account for nearly all of the cumulative excess: ages 60–69 (roughly 4.97 of the 7.73 excess deaths per 1,000, driven largely by NSHD) and ages 0–9 (0.70 per 1,000, driven by BCS70). As noted in Section 5.2, neither of these intervals has a straightforward panel-conditioning interpretation. The ages 60–69 estimate shrinks toward zero under narrow-window robustness checks, and the ages 0–9 estimate is mechanically implausible as a conditioning effect. Under the narrow-window specification (R1), the implied life-expectancy difference is correspondingly smaller and statistically indistinguishable from zero.

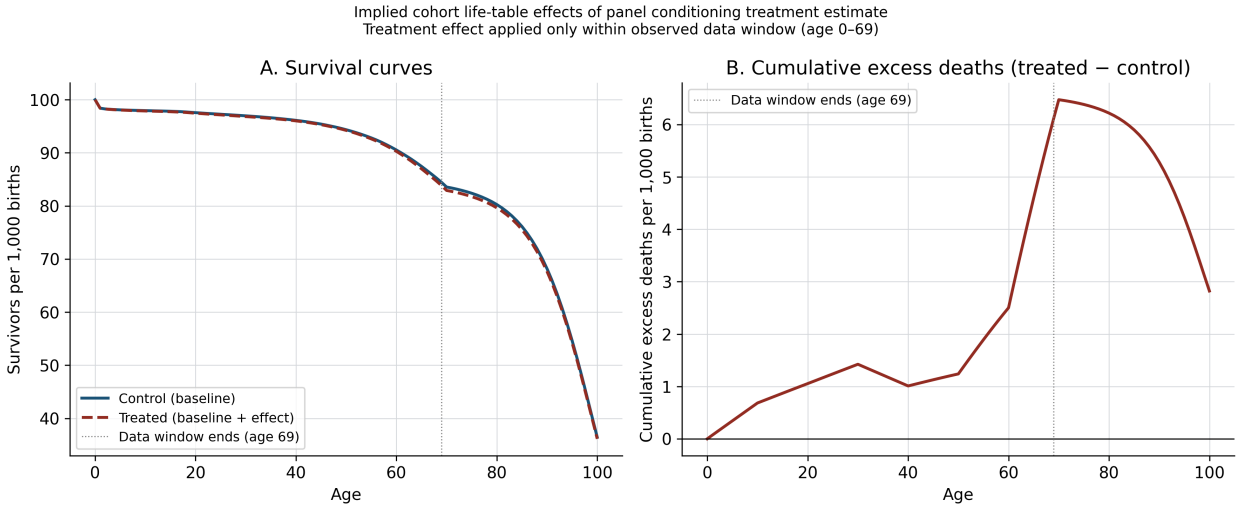

Figure 7: Implied cohort survival curves (Panel A) and cumulative excess deaths per 1,000 births (Panel B) under the baseline treatment-effect estimates. The dashed vertical line marks the end of the observed data window; beyond that point, baseline and treated life tables use identical mortality rates. These figures should be read as illustrative only.

8.2 B. Permutation test details

The permutation test rotates the treated indicator to each of the eight alternative week positions (1–7 and 9) within the target birth years (1946, 1958, 1970), keeping the same three-group pooled baseline specification. Each rotation yields a placebo coefficient from a world in which a different week had been selected by the cohort study. The eight placebo coefficients span a range from -0.13 to 0.07; the true estimate (0.060) is the second highest value. The one-sided permutation p-value — the proportion of values (including the truth) at least as large as the truth — is 0.222.

This finding does not reject the baseline result, but it does clarify its evidential weight: given only nine possible labelling arrangements (the selected position plus eight alternatives), even a truly null treatment effect would produce an estimate in the top two values roughly $2/9 = 22\%$ of the time by chance.

8.3 C. Validation against the Human Mortality Database

To assess whether our imputed mortality rates are consistent with independently measured national statistics, we compare the observed control-week mortality rates (averaged over birth years within each cluster) with the corresponding period death rates from the Human Mortality Database (HMD) for England and Wales (population code GBRTENW). HMD death rates are expressed in deaths per person per year and are multiplied by 1,000 to match our units. Crucially, we match each ONS observation to the HMD rate at the same single year of age and calendar year of death, rather than averaging HMD over the full window; this ensures the comparison reflects the same age–period combination in both series.

Table 7: Appendix Table A1: ONS control-week mortality rates vs. HMD period rates for England and Wales (GBRTENW), matched by age and calendar year of death. Both series in deaths per 1,000 per year. Ratios close to 1 indicate strong agreement; systematic values below 1 are consistent with our imputed denominator being modestly overstated relative to actual birth-week cohort size.

	Study	Age interval	ONS mean (SD)	HMD mean (SD)	ONS/HMD
0	NSHD 1946	20–29	0.647 (0.218)	0.676 (0.038)	0.96
1	NSHD 1946	30–39	0.873 (0.296)	0.962 (0.185)	0.91
2	NSHD 1946	40–49	1.937 (0.608)	2.192 (0.540)	0.88
3	NSHD 1946	50–59	4.330 (1.110)	5.014 (1.099)	0.86
4	NSHD 1946	60–69	7.579 (1.454)	9.654 (1.867)	0.79
5	NCDS 1958	10–19	0.487 (0.271)	0.508 (0.204)	0.96
6	NCDS 1958	20–29	0.596 (0.211)	0.615 (0.055)	0.97
7	NCDS 1958	30–39	0.862 (0.289)	0.924 (0.179)	0.93
8	NCDS 1958	40–49	1.789 (0.518)	1.941 (0.441)	0.92
9	NCDS 1958	50–59	3.154 (0.633)	3.505 (0.547)	0.90
10	BCS70 1970	0–9	1.470 (3.795)	1.542 (4.208)	0.95
11	BCS70 1970	10–19	0.348 (0.214)	0.370 (0.160)	0.94

Table 7: Appendix Table A1: ONS control-week mortality rates vs. HMD period rates for England and Wales (GBRTENW), matched by age and calendar year of death. Both series in deaths per 1,000 per year. Ratios close to 1 indicate strong agreement; systematic values below 1 are consistent with our imputed denominator being modestly overstated relative to actual birth-week cohort size.

	Study	Age interval	ONS mean (SD)	HMD mean (SD)	ONS/HMD
12	BCS70 1970	20–29	0.614 (0.212)	0.623 (0.045)	0.99
13	BCS70 1970	30–39	0.907 (0.269)	0.872 (0.141)	1.04
14	BCS70 1970	40–49	1.397 (0.370)	1.367 (0.151)	1.02

The ONS rates are 79–99% of the matched HMD values across all clusters and age intervals, with ratios closest to 1 at younger ages (0–29) and slightly below 1 at older ages. The systematic downward deviation is consistent with our fixed denominator ($N_b \approx 15{,}334$) being modestly too large for some birth years, or with secular cohort-survival advantages for the 1946–1970 birth years relative to earlier cohorts whose deaths also enter some HMD period cells. The key implication is that our rates are in the correct order of magnitude and their age trajectory matches the national benchmark; any remaining denominator error is approximately proportional and therefore absorbed by the birth-week fixed effects in the regression.